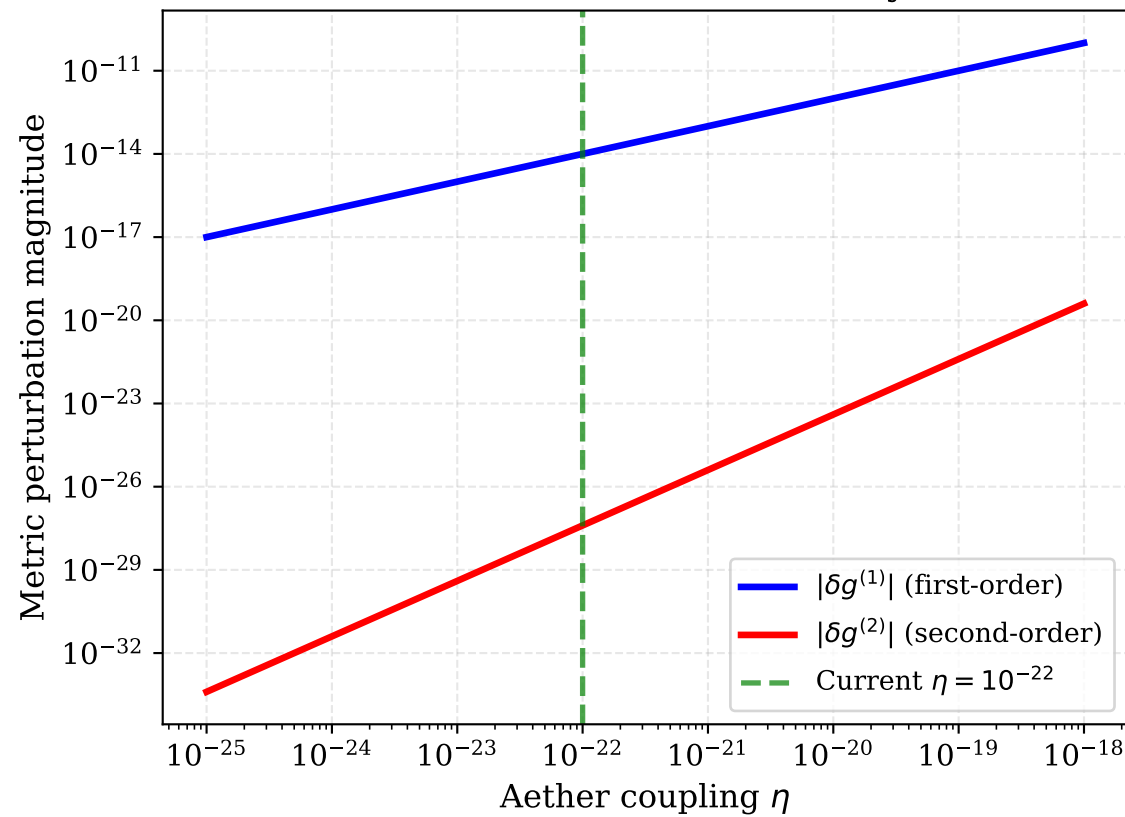
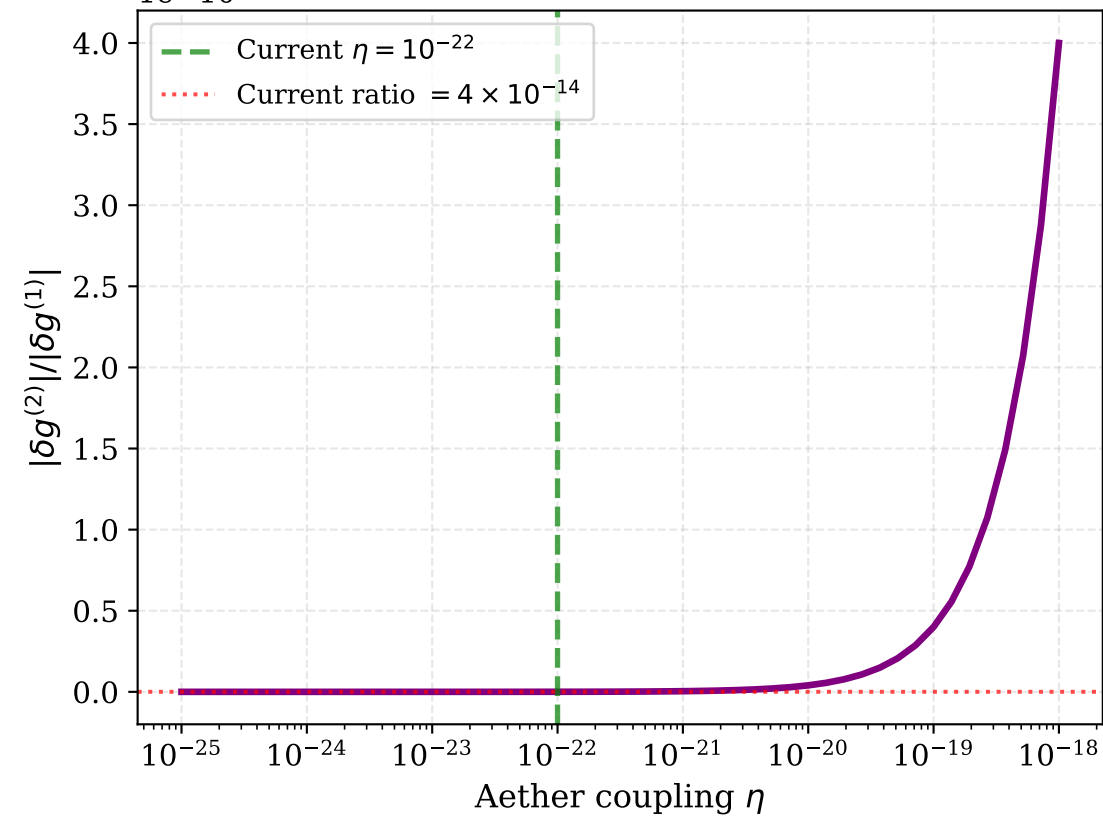
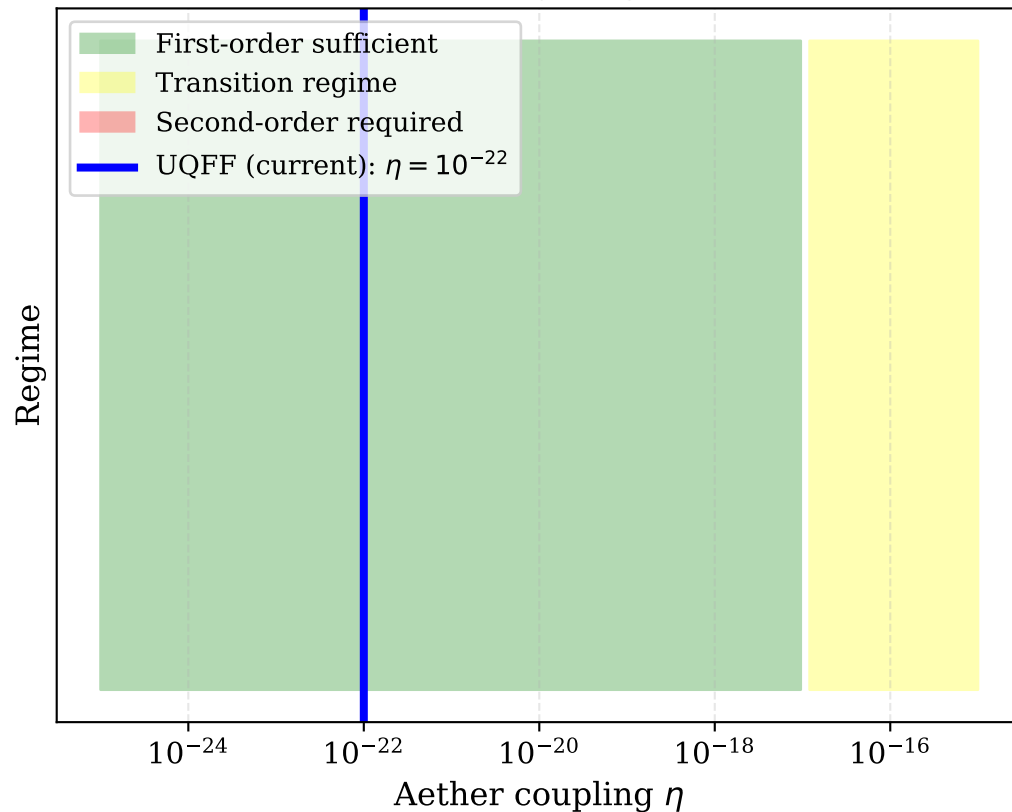


(a) Perturbation Hierarchy**(b) Second-Order to First-Order Ratio****(c) Validity Regimes****Higher-Order GR Corrections Summary****Metric Expansion:**

$$g_{\mu\nu} = \eta_{\mu\nu} + \varepsilon \delta g^{(1)} + \varepsilon^2 \delta g^{(2)} + O(\varepsilon^3)$$

Current UQFF ($\eta = 10^{-22}$):

- $|\delta g^{(1)}| \sim 10^{-14}$
- $|\delta g^{(2)}| \sim 4 \times 10^{-28}$
- Ratio: $4 \times 10^{-14} \ll 1$

Observational Precision:

- GPS timing: $\sim 10^{-12}$ (far above $\delta g^{(2)}$)
- LIGO: $\sim 10^{-21}$ (still above $\delta g^{(2)}$)
- Future quantum clocks: $\sim 10^{-18}$ (approaching)

Conclusion:

First-order treatment sufficient for all current and near-future observations. Second-order becomes relevant only if η increases to $\sim 10^{-10}$ or in ultra-strong field regimes.

Note: Higher-order corrections contain nonlinear terms: $(\delta g^{(1)})^2$ and $(\partial T_s)^2$.